ECA0906-DIGITAL SIGNAL PROCESSING

EXPERIMENT 10-DIGITAL SPEECH PROCESSING AND NOISE REDUCTION.

10.1 Implementation of Spectral Subtraction for Speech Noise Reduction.

Objectives:

1 To generate a noisy speech signal by adding noise to a clean speech signal.

2 To estimate the noise spectrum and implement the Spectral Subtraction algorithm.

3 To obtain and analyze the magnitude spectra of the noisy and enhanced speech signals.

4 To evaluate the performance of spectral subtraction in speech enhancement.

PROGRAM

```
clc;

clear;

close all;

%% Parameters

Fs = 8000;

t = 0:1/Fs:3;

%% Clean Signal

x = sin(2*pi*200*t) + 0.5*sin(2*pi*400*t);

%% Add Noise

rng(1);                   % For repeatable results
```

```
noise = 0.3*randn(size(x));

xn = x + noise;                  % Noisy signal

%% FFT of Noisy Signal

Y = fft(xn);

%% Estimate Noise

N = fft(noise);

%% Spectral Subtraction

magnitude_Y = abs(Y);

magnitude_N = abs(N);

% Subtract noise magnitude

magnitude_E = magnitude_Y - magnitude_N;

% Prevent negative values

magnitude_E = max(magnitude_E,0);

% Preserve phase information

phase_Y = angle(Y);

% Enhanced spectrum

Ye = magnitude_E .* exp(1j*phase_Y);

% Enhanced time-domain signal

xe = real(ifft(Ye));

%% ==================================================

%  TIME DOMAIN SIGNALS

% ==================================================
```

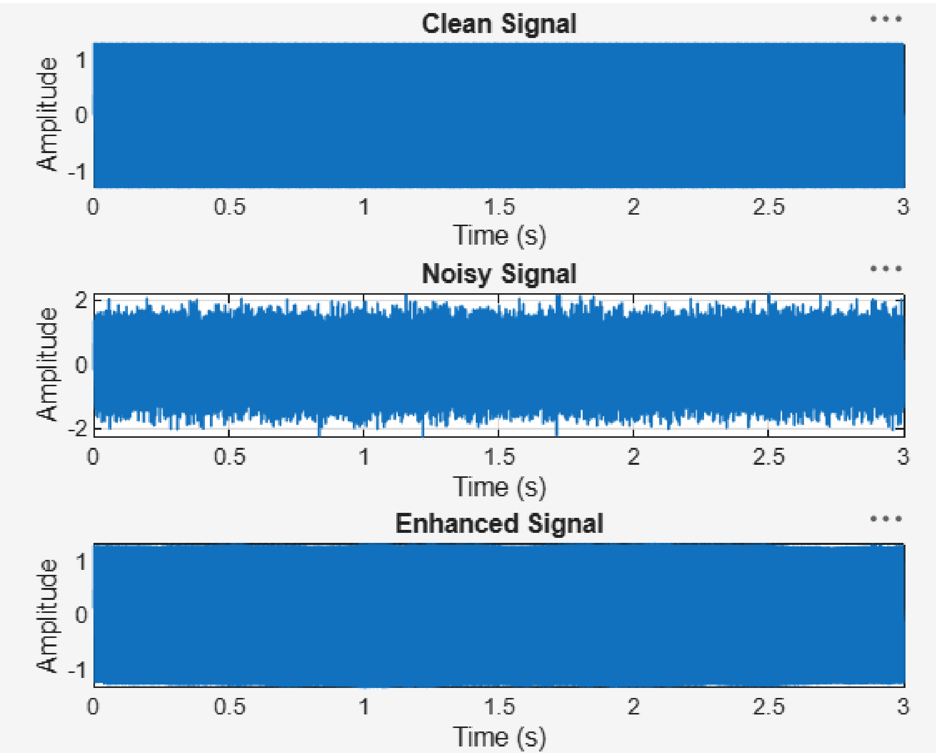
```
figure;
subplot(3,1,1);
plot(t,x,'LineWidth',1);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Clean Signal');
subplot(3,1,2);
plot(t,xn,'LineWidth',1);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Noisy Signal');
subplot(3,1,3);
plot(t,xe,'LineWidth',1);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Enhanced Signal');
%% ============================================================
% SPECTROGRAMS
% ============================================================
```

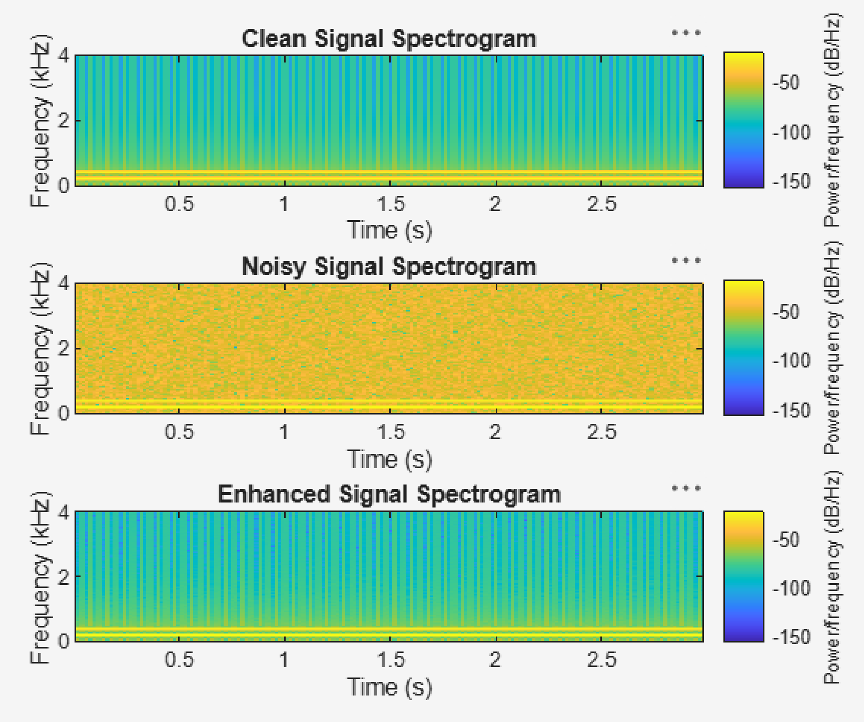
```
figure;
subplot(3,1,1);
spectrogram(x,256,128,256,Fs,'yaxis');
title('Clean Signal Spectrogram');
subplot(3,1,2);
spectrogram(xn,256,128,256,Fs,'yaxis');
title('Noisy Signal Spectrogram');
subplot(3,1,3);
spectrogram(xe,256,128,256,Fs,'yaxis');
title('Enhanced Signal Spectrogram');
%% ==========================================================
% MAGNITUDE SPECTRUM
% ==========================================================
L = length(x);
f = (0:L-1)*(Fs/L);
X = abs(fft(x))/L;
XN = abs(fft(xn))/L;
XE = abs(fft(xe))/L;
%% Plot only first half of spectrum
figure;
subplot(3,1,1);
plot(f(1:L/2),X(1:L/2),'LineWidth',1);
```

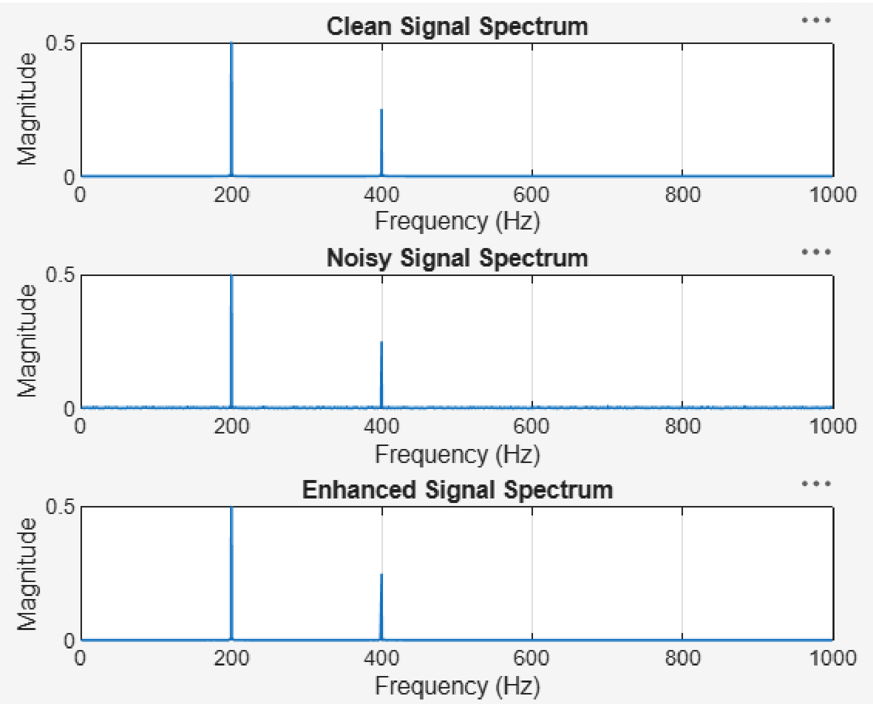
```
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Clean Signal Spectrum');
xlim([0 1000]);
subplot(3,1,2);
plot(f(1:L/2),XN(1:L/2),'LineWidth',1);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Noisy Signal Spectrum');
xlim([0 1000]);
subplot(3,1,3);
plot(f(1:L/2),XE(1:L/2),'LineWidth',1);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Enhanced Signal Spectrum');
xlim([0 1000]);
%% ==================================================
% PERFORMANCE COMPARISON
% ==================================================
```

```
mse_noisy = mean((x-xn).^2);

mse_enhanced = mean((x-xe).^2);

fprintf('\n===================================\n');

fprintf(' SPEECH ENHANCEMENT RESULTS\n');

fprintf('===================================\n');

fprintf('MSE of Noisy Signal    = %.6f\n',mse_noisy);

fprintf('MSE of Enhanced Signal = %.6f\n',mse_enhanced);

fprintf('===================================\n');
```

OUTPUT







RESULT

A noisy speech signal was successfully generated by adding noise to the clean speech signal.The noise spectrum was estimated and the Spectral Subtraction algorithm was successfully implemented.